

Where is the gender gap in generative AI use widest?

Evidence across European countries, age groups, and use contexts

Emily Birch · Research draft · September 2026

In the European Union, 30.45% of women and 34.91% of men reported using generative AI during the three months before their 2025 survey interview. That leaves a gap of **4.46 percentage points**. But the European average gives us little sense of where the difference is most pronounced, or whether it appears in the same way across everyday life, work and education.

Looking more closely changes the picture. **Ireland has the largest observed country gap in percentage points**, while **Poland has the largest proportional shortfall in women's use**. Among the youngest adults, women report more use than men at EU level. And the difference in private use is larger than the difference in work use. These patterns matter for how we describe participation in a technology whose social and economic effects we are trying to understand.

This study asks **where those gaps are largest, smallest or reversed**, and **how much the answer changes when we account for age composition or measure the gap differently**. I use Eurostat's [2025 survey estimates](#) for people aged 16-74, comparing countries, age groups, educational attainment and purposes of use. The analysis describes participation; it does not try to establish why the differences arise.

The European scope makes a consistent comparison possible. Eurostat provides a [harmonised survey framework](#) across countries, avoiding many of the differences in definitions and populations that would complicate a worldwide ranking assembled from unrelated surveys. The main analysis covers all 27 EU countries, with eight additional countries and territories reported separately in the supporting tables.

Throughout, **“use” means reported use in the preceding three months**, following the [survey questionnaire](#). Someone who tried a tool once and someone who uses it daily can both count as users. These estimates therefore tell us about recent participation, rather than frequency, skill or benefit. The survey records male and female categories, so the comparisons also cannot represent the full range of gender identities.

The first comparison is between countries. Figure 1 shows women's and men's use rates side by side: the distance between them is the gap in percentage points. I include a comparison only when both estimates are available and neither carries a reliability flag. The public extract does not provide the information needed to calculate confidence intervals for these gaps, so **“largest” means the largest published estimate**, rather than a statistically established first place.

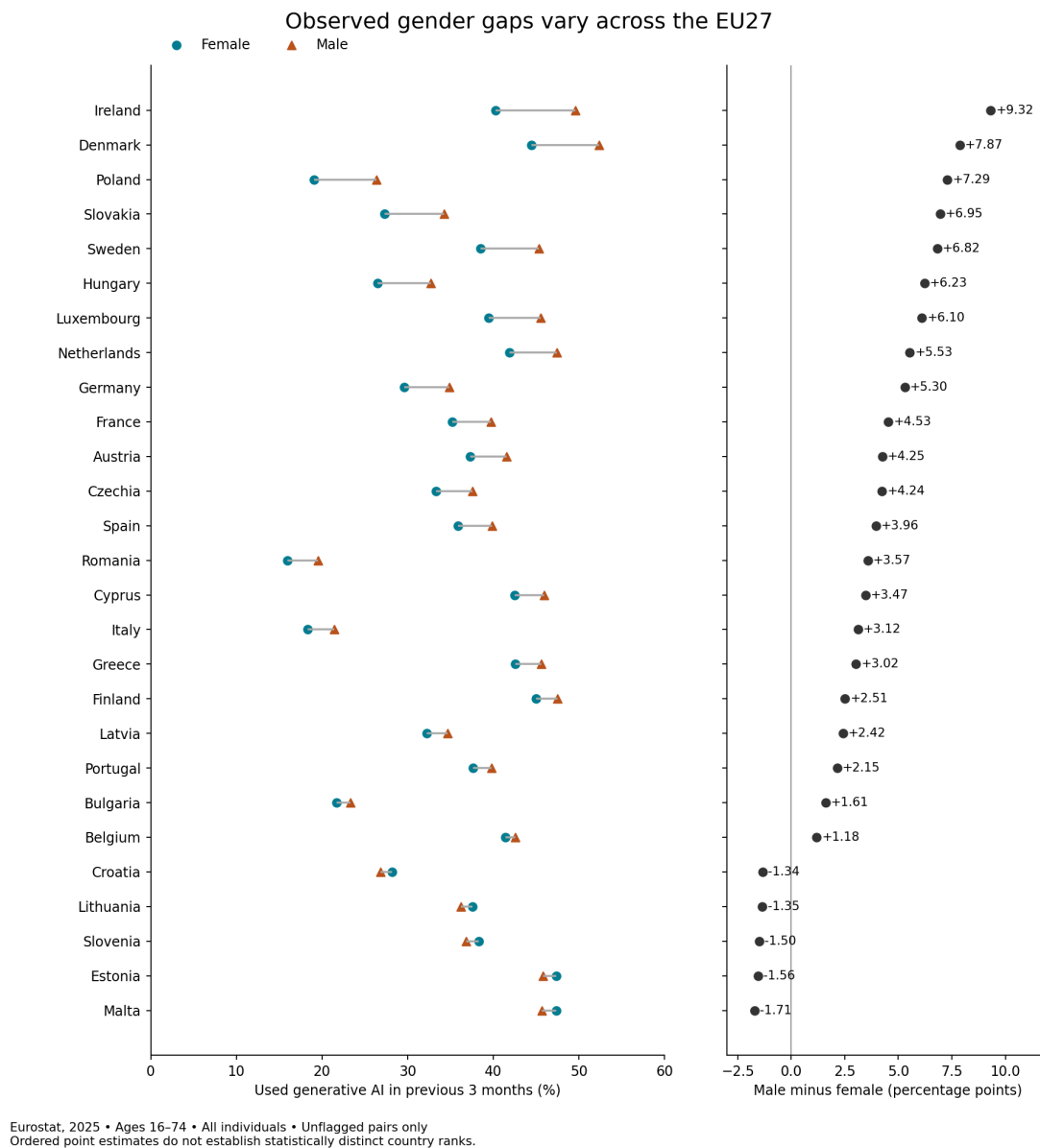


Figure 1. Overall use among people aged 16-74, as a percentage of all individuals. Positive gaps indicate higher male use. All country pairs are unflagged. [Underlying table](#).

Ireland has the largest observed gap: 9.32 percentage points, with use reported by 40.30% of women and 49.62% of men. Denmark follows at 7.87 points and Poland at 7.29. Yet Denmark and Poland have very different levels of participation. In Denmark, 44.49% of women report use; in Poland, the figure is 19.10%. A relatively large gap can occur where AI use is widespread as well as where it is less common.

This difference in underlying participation helps explain why the country with the widest gap depends on how we measure it. In Poland, women's use is approximately 27.6% below men's: the female rate is 0.724 times the male rate. In Ireland, the corresponding ratio is 0.812. Ireland has the larger distance between the two rates, but Poland has the larger shortfall relative to men's use.

Both comparisons are useful. **Percentage points** tell us how far apart women's and men's participation rates are. **The ratio** tells us how common use is among women relative to men.

Reporting the two rates alongside the gap makes it possible to see both, rather than letting a single country ranking stand in for the whole picture.

Male use is also not higher in every country. Malta, Estonia, Slovenia, Lithuania and Croatia have higher female estimates, with differences ranging from 1.34 to 1.71 percentage points. Those differences are small enough that sampling uncertainty is particularly relevant: we cannot establish that women in these populations are more likely to use AI. The estimates nevertheless show that the observed country pattern is not uniformly a male advantage.

Country totals combine people at very different stages of life. Splitting the same comparison into age groups shows a further pattern hidden by the European average.

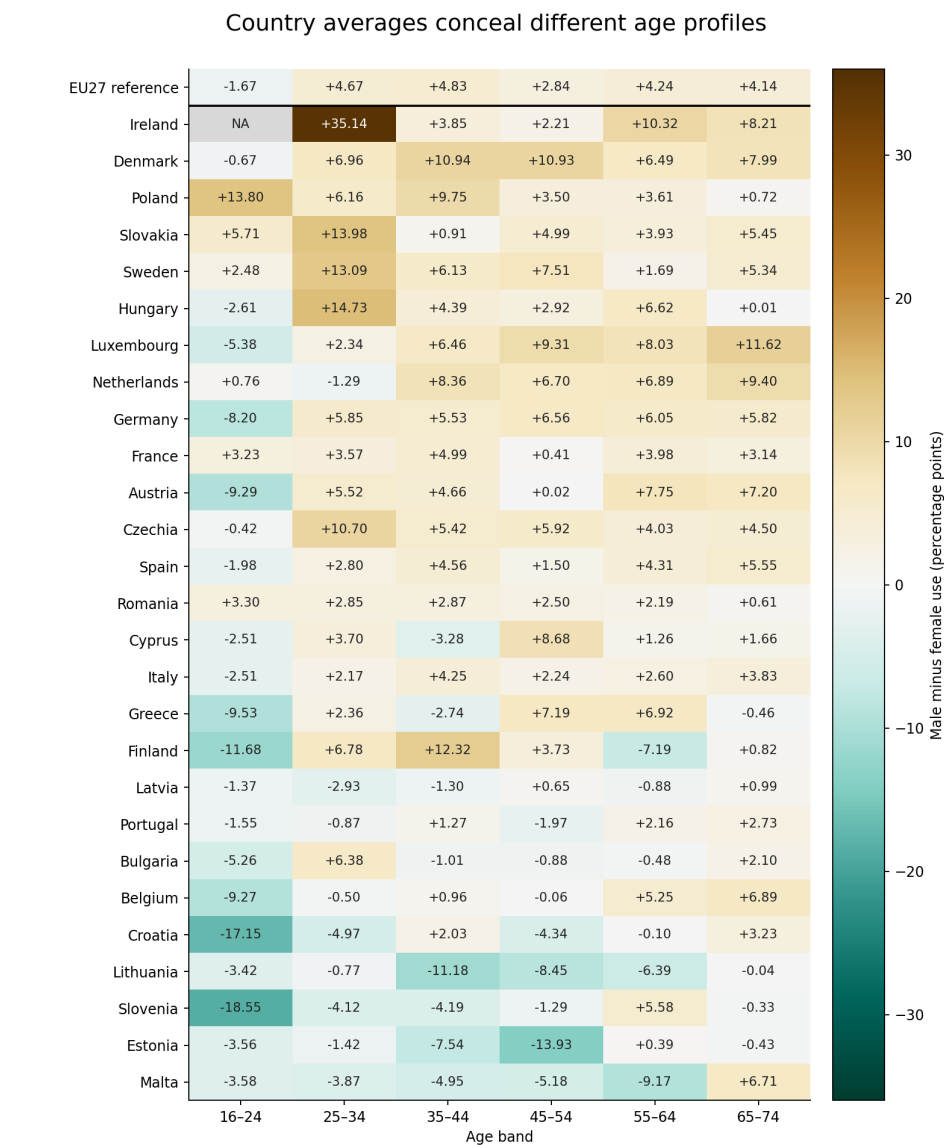


Figure 2. Recent use by six age bands. Each cell is male minus female use in percentage points. Countries appear in the same order as Figure 1. Grey cells have a missing or flagged estimate. [Underlying table](#).

At EU level, **64.67% of women aged 16-24 report use, compared with 63.00% of men** of the same age. Men have the higher estimate in each of the five older age groups, with the largest percentage-point gap at ages 35-44: 4.83 points.

The oldest group illustrates why an absolute gap alone can miss an important difference. Among people aged 65-74, use is reported by 4.58% of women and 8.72% of men. The 4.14-point gap is smaller than the gap among those aged 35-44, yet **women's use is only about half as common as men's**. When overall use is low, a difference of a few percentage points can represent a substantial proportional disparity.

Figure 2 also shows why the EU average should not be treated as a rule for every country: the direction and size of the gap vary within the same age band. These finer comparisons deserve more caution than national totals, since subgroup estimates can be less reliable even when they carry no flag. Ireland, for example, has an unflagged overall estimate but a low-reliability estimate for its youngest male group. That group is excluded from the main comparisons.

The youngest group's pattern raises an interesting question about the future, but this survey cannot tell us whether the overall gap will shrink as younger people grow older. We observe different age groups at one point in time, rather than following the same people as they age. Their differences could reflect life stage, the experiences of different generations, or both.

Educational attainment adds another dimension. The EU percentage-point gap is 3.76 among people with low attainment, 4.61 among those with medium attainment and 6.97 among those with high attainment. The absolute difference is therefore **largest in the most highly educated group**. That does not show that education itself increases the gap: the groups also differ in age and other circumstances. The available tables do not allow age and education to be adjusted together. [Education results](#).

So far, the comparisons have counted any recent use. Asking what people use AI for brings us closer to the different settings in which participation might matter. The [Eurostat questionnaire](#) distinguishes private, work and formal-education use, and **respondents can report more than one purpose**.

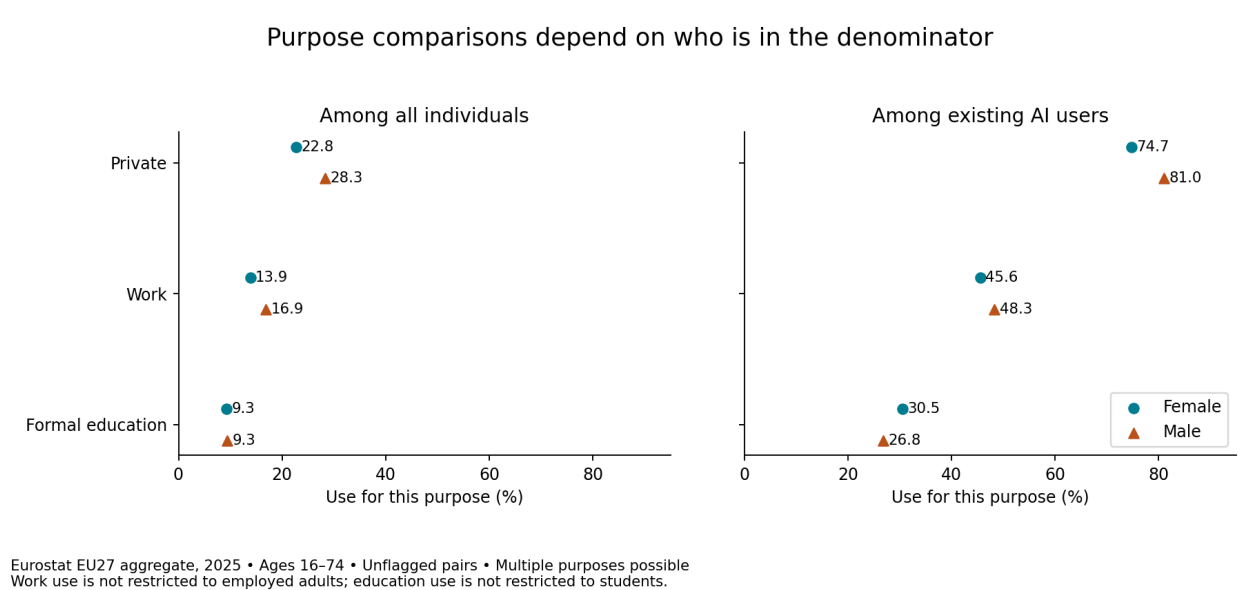


Figure 3. The left panel asks how common each purpose is among all women or men. The right asks how common it is among women or men who already use AI. Purposes can overlap. [Underlying table](#).

The left panel shows **a larger gender gap in private use than in work use**: 5.51 percentage points compared with 3.00. **Formal-education use is almost equal**, reported by 9.30% of women and 9.35% of men. The aggregate gender gap therefore looks quite different across these three contexts.

The private-work comparison also appears across much of the EU. In 23 of the 27 countries, subtracting women's use from men's gives a higher value for private use than for work use. This counts each country once and preserves the direction of each difference; it does not mean that private use has the larger disparity regardless of which sex reports more use. The [country-purpose table](#) reports both rates so these comparisons can be inspected directly.

These broad categories need careful interpretation. The work-use figures include all people aged 16-74, including those without jobs. A difference could therefore reflect who is employed and which jobs they do, as well as differences in use among people doing comparable work. Similarly, private use can include activities such as household administration as well as entertainment. The category alone does not tell us the economic or social value of that use.

The education result becomes clearer when we distinguish two questions. **How many women and men use AI for education?** And **among those already using AI, how many report education as a purpose?**

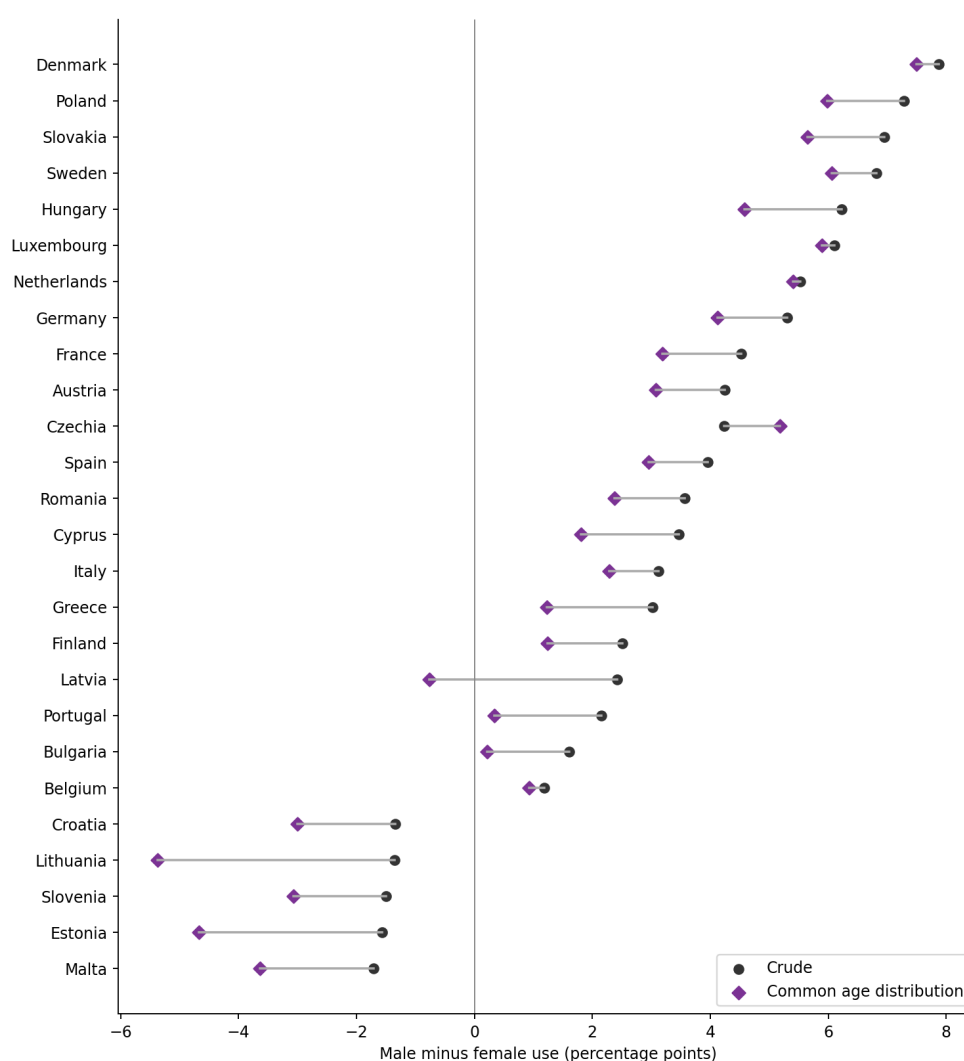
For the first question, the rates are nearly identical: around nine in every hundred women and nine in every hundred men. But **fewer women report any AI use in the first place**. Those reporting education use therefore make up a larger proportion of the female users. Among AI users, 30.53% of women report education use, compared with 26.77% of men, as the right panel shows.

This is why similar participation rates in the whole population can coexist with different patterns among users. It does not tell us that female students are more likely to use AI than male students: neither calculation is restricted to students. Nor does it measure how much time people spend using AI for education. It tells us how many people report that purpose, using two different reference populations.

Age differences raise one final question about the country comparison. **Would the gaps look the same if women and men in every country had the same age distribution?** This matters because country totals reflect both the use rates within each age group and the proportion of people in that group.

To make that comparison, I keep each country's age-specific female and male use rates, but give each age group the same weight everywhere. Those weights reflect the [combined female and male EU27 population aged 16-74](#). For example, the 65-74 group contributes the same proportion to every country's recalculated female and male rate. This procedure, called **age standardisation**, lets us compare the gaps under a common age distribution.

Age standardisation changes some observed gaps substantially



Eurostat, 2025 • Overall use, ages 16–74 • All individuals • Same 26 EU countries in both series
Ireland excluded: incomplete reliable age pairs. Weights: pooled EU27 resident population, 1 January 2025 (ep).
Descriptive adjustment; no causal attribution or sampling uncertainty estimated.

Figure 4. Unadjusted and age-standardised gaps, using the same six age-group weights for women and men in every country. The weights come from estimated/provisional EU27 resident population on 1 January 2025. Ireland is excluded because its youngest male estimate is flagged. [Underlying table](#).

The comparison covers 26 EU countries with complete, unflagged age estimates. Across them, the median change in the gap, regardless of direction, is 1.31 percentage points. Some country results change appreciably. Lithuania's estimated female advantage grows from 1.35 to 5.38 points. **Latvia moves from a 2.42-point male advantage to a 0.77-point female advantage.**

Latvia illustrates how the direction of an overall comparison can depend on the age mix used to calculate it. It does not establish a statistically significant reversal, or identify how much of a gender gap is caused by age. The exercise changes the weights used to combine age groups; other differences between women and men remain. The [methods appendix](#) explains the calculation and its limitations, including the use of resident-population weights for a household survey.

Other checks examine whether the findings depend on who is included. Restricting the comparison to recent internet users changes the sign of Lithuania's overall gap, even without age standardisation.

Allowing estimates flagged for low reliability adds some subgroup comparisons, but leaves the overall country and purpose comparisons unchanged. These checks are reported in the appendix so the main results can be read alongside the consequences of different choices.

The analysis adds detail to an existing body of research. Humlum and Vestergaard document [unequal workplace adoption of ChatGPT in Denmark](#), while Henseke's [study of generative AI in European workplaces](#) maps country-specific gender gaps using a worker survey. The [global synthesis by Cranney, Delecourt and Koning](#) already includes Eurostat's EU aggregate. The contribution here is to **examine variation within the harmonised household survey** and show how the description changes with age composition, purpose and measurement. The [literature notes](#) discuss further related work and differences between the studies' populations and definitions.

Harmonisation does not remove every difference between national surveys. Fieldwork took place at different times during 2025; [Greece's survey](#), for example, ran from July to September. In a period of changing use, the country comparison is therefore not a snapshot of one shared calendar window. The [country register](#) records the available information on timing and implementation. Respondents may also differ in whether they recognise an activity as generative AI use. These limitations, together with the absence of sampling uncertainty for the gaps, place boundaries on how finely we can interpret the results.

What emerges is a picture of uneven participation that the European average cannot capture. Country rankings depend on whether we ask about absolute or proportional differences. The age profile includes both a female advantage among the youngest adults and a substantial proportional shortfall among the oldest. Private, work and education use show different patterns, and some country comparisons change when age composition is held constant.

For research on AI's social and economic consequences, this is a useful starting point: it identifies where participation differs and which comparisons deserve closer investigation. **Whether those differences translate into unequal opportunities or benefits remains an open question.** Answering it requires evidence about what people do with AI and what follows from that use, beyond whether they used it recently.

Data and methods. The [abstract](#), [methods appendix](#), [references](#), [source register](#), [code](#) and [numerical claims table](#) accompany this draft. The data were frozen on 21 September 2026. The analysis plan followed disclosed exploratory inspection and was not preregistered before seeing the data. AI assistance supported source retrieval, coding, verification and drafting. This draft awaits author review and has not received independent peer review.